

Konsider Phase 6B Prompt — Career Opportunity Criteria and Evidence Exploration

Role

Work as the lead product, data-research, and evidence engineer for the Konsider repository.

This is a research and product-definition phase only.

Do not implement runtime schemas, production workers, ranking changes, release changes, API changes, UI changes, or preference presets.

Temporary research scripts, retained local source captures, and committed research reports are allowed where necessary for deterministic analysis.

Background

Konsider currently has:

- a stable universe of 91 countries;
- 13 enabled ordering criteria that may affect affinity scores;
- full-coverage and partial-coverage ranking rules;
- complete-case ranking with no imputation;
- structured coverage, locality, and profile assessments; and
- no applicant-specific career assessment.

The earlier Phase 6A study attempted to publish occupation and education criteria as weightable ranking criteria. It approved none of the seven candidates.

The occupation-family evidence was meaningful but covered only 61–66 fresh countries. It therefore failed the existing minimum of 82 valid countries and the combined PCC missing-union limit.

Do not repeat Phase 6A mechanically. This phase evaluates a different product concept.

New product concept: Opportunity Criteria

An **Opportunity Criterion** is a non-ranking, tri-state filter.

It does not:

- contribute to affinity score;
- change the ordering of eligible countries;
- impute values;

- interpret missing evidence as poor opportunity; or
- estimate a particular applicant's probability of employment.

It may allow a user to request:

Only consider countries where Konsider has verified a strong opportunity signal for my chosen career field.

The existing ordering criteria rank the countries that remain after the filter.

Required public states

Every country must have exactly one of these states for every evaluated Opportunity Criterion:

VERIFIED_STRONG_SIGNAL

Reproducible evidence crosses the frozen threshold for a substantial destination-side opportunity ecosystem.

STRONG_SIGNAL_NOT_ESTABLISHED

The country has sufficient, comparable evidence to assess the opportunity, but the evidence does not cross the strong-signal threshold.

This does not mean that no jobs or opportunities exist.

INSUFFICIENT_EVIDENCE

The available evidence is missing, stale, materially incomplete, incomparable, contradictory, legally unusable, or otherwise insufficient for either of the first two outcomes.

This state must never be presented as a negative assessment.

Asymmetric evidence rule

Positive and negative evidence need not require identical source completeness.

A country may receive **VERIFIED_STRONG_SIGNAL** when one approved high-confidence evidence path, or a defined combination of approved evidence paths, establishes a strong ecosystem.

A country may receive **STRONG_SIGNAL_NOT_ESTABLISHED** only when the research demonstrates that the evidence process had a reasonable opportunity to detect a strong ecosystem and did not detect one.

When important evidence paths are unavailable, the country must remain **INSUFFICIENT_EVIDENCE**.

Do not infer a negative result merely because:

- ILOSTAT lacks a row;
- a company list does not mention the country;
- a web search returns few results;
- the country has no headquarters from a selected company list; or
- one source family has weak coverage.

Repository orientation

Read at minimum:

- docs/research/phase6a-career-education-source-study.md
- data/reports/phase6a-2026-07-30/decision-matrix.json
- data/reports/phase6a-2026-07-30/country-coverage.jsonl
- data/reports/phase6a-2026-07-30/coverage-summary.json
- data/reports/phase6a-2026-07-30/sources.json
- docs/history/phase5-closure-report.md
- docs/history/phase4-closure-report.md
- docs/research/phase3-closure-report.md
- the active release report;
- the current catalog and preference presets; and
- the current source, scope, applicability, and lineage contracts.

Treat the current repository as authoritative where it has moved beyond these documents.

Objective

Identify a small, user-comprehensible portfolio of career Opportunity Criteria for which strong, reproducible positive and negative signals can be established for a practically useful portion of the 91-country universe.

The phase must answer:

1. Which career fields are important enough to deserve a public Opportunity Criterion?
2. What exactly does “strong opportunity” mean for each field?
3. Which evidence families can establish that signal?
4. Can at least 60 of the 91 countries be assessed?
5. Are the countries most likely to appear in user shortlists adequately covered?
6. Can positive, negative, and insufficient-evidence states be assigned honestly?
7. Which candidates should proceed to a future implementation-design phase?

Do not implement the Opportunity Criteria in this phase.

Starting candidate portfolio

Research the following as candidate user-facing Opportunity Criteria.

The final names may be narrowed when source granularity requires it.

Core candidates

1. Job opportunities — Technology and software

- 2. Intended meaning: a substantial and established ICT/software employment ecosystem.
- 3. Do not imply live vacancies, salary, visa access, language fit, seniority fit, or personal hiring probability.

4. Job opportunities — Science and engineering

- 5. Intended meaning: a substantial employment ecosystem in science, engineering, and related technical occupations.
- 6. Narrow or rename if evidence supports only specific occupation groups.
- 7. Do not call broad science-and-engineering evidence “pure engineering.”

8. Job opportunities — Healthcare

- 9. Intended meaning: a substantial healthcare employment ecosystem.
- 10. Do not imply that a foreign doctor, nurse, or health professional is licensed or employable.

11. Job opportunities — Business, finance and administration

- 12. Intended meaning: a substantial professional employment ecosystem in defensibly mapped business, finance, accounting, and administrative fields.
- 13. Avoid arbitrary mixing of management, sales, legal, clerical, and finance occupations.
- 14. Narrow the public name if the evidence supports only business and administration.

Additional candidates to probe

1. Job opportunities — Research and academia

- 2. Intended meaning: a substantial research, university, or research-institution employment ecosystem.
- 3. Do not imply open academic positions or applicant eligibility.

4. Job opportunities — Skilled trades and construction

- 5. Intended meaning: a substantial destination-side skilled-trades and construction employment ecosystem.

6. Treat professional certification and licensing as future applicant-context questions.

Do not add more public candidates merely because source codes exist. Prefer a small number of broad, understandable fields over many thin occupation filters.

Construct discipline

For each candidate, freeze a precise destination-side question.

Example:

Does this country have a sufficiently substantial and established technology employment ecosystem to constitute a verified strong opportunity signal?

The construct must not become:

- likelihood of getting hired;
- count of jobs currently open;
- future employment growth unless supported by comparable forecasts;
- salary or affordability;
- visa or residence eligibility;
- qualification recognition;
- professional licensing;
- city-specific access without locality evidence; or
- overall country quality.

Evidence-family exploration

Do not assume that one source must cover every country.

Research whether a controlled combination of evidence families can establish the tri-state outcome.

A. Harmonised occupation evidence

Start from the Phase 6A ILOSTAT work.

Evaluate:

- occupation-family employment stock;
- occupation-family share of total employment;
- log-adjusted ecosystem scale;
- compatible denominator and period;
- national-source comparability;
- classification level;
- observed versus modelled status;
- breaks and unreliable flags;

- newer editions or alternate official ILOSTAT distributions;
- whether any other official international source materially improves coverage.

Do not lower freshness or accept unreliable observations merely to increase coverage.

B. Major-employer ecosystem evidence

Explore whether a reproducible employer-presence signal can supplement occupation statistics.

Possible evidence may include:

- official company office or engineering-centre directories;
- annual reports;
- official workforce-location disclosures;
- official subsidiary or operational-location registries;
- authoritative global company lists with acceptable reuse;
- multiple major employers rather than one company;
- country-level headcount only where explicitly published and comparable.

Reject simplistic office counting where a small sales office is treated as equivalent to a large engineering or operational centre.

Do not rely on search-result counts, crowdsourced office lists, or manually curated claims without reproducible source identity.

C. Current-demand evidence

Explore official or authoritative vacancy, shortage, or employment-demand evidence only where cross-country comparability is credible.

Possible evidence families may include:

- official national vacancy statistics;
- public employment-service datasets;
- internationally harmonised vacancy indicators;
- official shortage-occupation lists;
- transparent commercial datasets with production-compatible licensing.

Treat shortage lists cautiously because they may reflect immigration policy rather than total labour-market depth.

Do not scrape commercial job sites unless legal reuse, stable access, reproducibility, and country comparability are demonstrated.

D. Supporting ecosystem evidence

Where relevant, explore:

- research activity;
- sector output;
- number and breadth of relevant institutions;
- startup or employer concentration;
- recognised industry clusters.

Supporting evidence must remain secondary unless it directly measures the approved construct.

Composite evidence design

For each candidate, evaluate simple, explainable decision rules.

Examples to test—not assumptions to adopt—include:

Route 1: direct strong signal

A harmonised occupation measure crosses both:

- a minimum ecosystem-scale threshold; and
- a minimum specialisation threshold.

Route 2: corroborated strong signal

Moderate occupation evidence plus one or more independently verified ecosystem signals, such as:

- broad major-employer presence;
- credible current-demand evidence; or
- substantial sector/institution breadth.

Negative assessment

`STRONG_SIGNAL_NOT_ESTABLISHED` may be assigned only when:

- the country has adequate comparable evidence under the approved detection framework;
- no strong-signal route passes; and
- major known evidence gaps are absent.

Otherwise assign `INSUFFICIENT_EVIDENCE`.

Do not create an opaque machine-learning score or a large composite with arbitrary weights.

Prefer:

- clear thresholds;
- a limited number of evidence routes;
- visible evidence contributions;
- frozen source precedence;
- explicit reason codes; and
- sensitivity analysis around thresholds.

Minimum research gate

A candidate may be recommended for implementation only if all of the following pass.

1. Assessable-country floor

At least **60 of the stable 91 countries** can be classified as either:

- VERIFIED_STRONG_SIGNAL ; or
- STRONG_SIGNAL_NOT_ESTABLISHED .

Countries classified INSUFFICIENT_EVIDENCE do not count.

2. Decision-relevance coverage

Using only the currently published ordering criteria, produce benchmark top-20 shortlists for at least:

- general balanced preferences;
- career-prioritised preferences;
- family and education-oriented preferences;
- safety and governance-oriented preferences; and
- affordability-sensitive preferences.

For every candidate, report evidence coverage for each shortlist.

The preferred implementation gate is at least **16 of 20 assessable countries in each relevant shortlist**.

A candidate below this level may still be retained as research-only, but the gap must be explicit.

Do not use the new opportunity evidence to generate these benchmark lists.

3. Geographic and economic breadth

Assessable evidence must not be almost entirely limited to:

- Europe and North America;
- high-income countries;
- one publisher family;

- one language group; or
- one regulatory system.

Report coverage by region and World Bank income group or another frozen development grouping.

4. Discrimination

The criterion must not be effectively saturated.

Report:

- number of strong-signal countries;
- number of assessed-no-strong-signal countries;
- number with insufficient evidence;
- regional distribution;
- sensitivity to threshold changes.

As a normal guide, the smaller assessed class should contain at least eight countries.

Do not force balance if the real evidence is asymmetric, but flag a candidate that provides little filtering value.

5. Construct integrity

The evidence and public name must match.

Examples:

- employment stock is not live job availability;
- major-company presence is not total sector employment;
- shortage-list presence is not guaranteed employability;
- research output is not job availability;
- occupation depth is not professional licensing access.

6. Source and legal gate

For every source path freeze:

- exact asset or API query;
- publisher and distributor;
- version or extraction date;
- terms and licence;
- permitted production use;
- permitted normalized derivative use;
- attribution;
- raw-retention policy;
- checksum or deterministic capture identity;
- refresh approach.

Ambiguous production rights block implementation recommendation.

7. Reproducibility

A fresh checkout with the approved retained inputs must be able to reproduce:

- source normalization;
- country mapping;
- evidence components;
- status assignment;
- reason codes;
- coverage analysis;
- shortlist analysis; and
- final decision matrix.

Country-level output

For every candidate and all 91 countries, produce a structured record containing at minimum:

- candidate ID;
- proposed public name;
- country code;
- public state;
- reason codes;
- evidence paths evaluated;
- evidence paths unavailable;
- exact source references;
- latest applicable period;
- positive threshold results;
- negative-assessment completeness result;
- contradictions or caveats;
- confidence tier;
- and whether the country belongs to any benchmark top-20 shortlist.

Do not assign production states based solely on researcher judgement. The record must follow a deterministic proposed policy.

Required analysis

For each candidate report:

- precise construct;
- user decision value;
- source families considered;
- exact source assets;
- legal conclusions;
- coverage by state;

- benchmark-shortlist coverage;
- region and income-group coverage;
- source-overlap and source-dependency;
- threshold sensitivity;
- strong-signal examples;
- assessed-no-strong-signal examples;
- insufficient-evidence examples;
- false-positive risks;
- false-negative risks;
- maintenance burden;
- refresh expectations;
- and the boundary with future applicant context.

Explicitly analyse prominent missing countries from Phase 6A, including where applicable:

- Canada;
- Japan;
- South Korea;
- New Zealand;
- Malaysia;
- South Africa; and
- other countries that frequently appear in benchmark shortlists.

Decision categories

Assign each candidate exactly one disposition:

APPROVE_FOR_IMPLEMENTATION_DESIGN

The candidate clears the research gates and has a sufficiently precise tri-state evidence policy.

This approves future model/API/UI design work. It does not publish the criterion.

APPROVE_WITH_NAMING_OR_SCOPE_CHANGE

The evidence supports a useful narrower construct, but not the original public name.

Freeze the recommended replacement name and meaning.

HOLD_SOURCE_GAP

The construct is useful, but source coverage, licensing, comparability, or reproducibility is insufficient.

State the exact reopening condition.

HOLD_POLICY_OR_THRESHOLD_GAP

Evidence exists, but the tri-state policy, threshold, negative-state completeness, or shortlist coverage is not yet defensible.

REJECT_AS_OPPORTUNITY_CRITERION

The candidate is too ambiguous, too thin, too redundant, too applicant-specific, or not useful as a public filter.

Rejection must be construct-specific and must not delete prior evidence.

Required deliverables

Create:

1. docs/research/phase6b-career-opportunity-study.md
2. data/reports/phase6b-<date>/career-candidate-matrix.json
3. data/reports/phase6b-<date>/career-country-opportunity-evidence.jsonl
4. data/reports/phase6b-<date>/career-source-matrix.json
5. data/reports/phase6b-<date>/career-shortlist-coverage.json
6. data/reports/phase6b-<date>/career-threshold-sensitivity.json
7. data/reports/phase6b-<date>/approved-career-opportunity-portfolio.json
8. any small deterministic research scripts and fixtures required to reproduce the analysis.

The approved portfolio file must contain only candidates recommended for a future implementation-design phase.

Explicit non-goals

Do not:

- change the active release;
- create a new production release;
- modify ranking logic;
- modify PCC policy;
- modify the stable 91-country universe;
- add runtime Opportunity Criterion schemas;
- add API routes;
- add UI controls;
- add conversational behavior;
- add applicant profiles;
- add visa or licensing logic;
- lower existing ranking-publication gates;
- or publish opportunity labels.

Verification

Run all repository gates relevant to changed research scripts and documentation.

At minimum:

- deterministic research replay;
- output checksum verification;
- JSON/JSONL validation;
- country-count reconciliation;
- no missing country-candidate rows;
- documentation-link checks;
- lint and formatting for added scripts;
- existing regression tests affected by documentation or research-path changes.

Do not claim remote CI unless it has actually run.

Stop condition

Stop after the research artifacts and decision report are complete.

Report:

- candidates researched;
- exact approved public names;
- assessable-country counts;
- top-20 shortlist coverage;
- strong/no-strong/insufficient counts;
- source and legal blockers;
- implementation-design approvals;
- held or rejected candidates;
- files changed;
- tests run;
- commit SHA;
- and decisions required from the owner.

Do not proceed to code, contracts, releases, API, or UI.